Digital Vector · VCE Applied Computing | Unit 2 · Area of Study 2 · Cyber Security · KK06

Risks & benefits of networks in a global environment

Above: PixelPort, run from a bedroom in Melbourne, with players, servers and a payment processor scattered across the globe. Spin the planet and switch on each layer — every line is something that has to stay safe and legal in more than one country at once.

What this key knowledge covers
This summary distils everything you need for **KK06 — Risks & benefits of networks in a global environment** in Cyber Security. Work through the explanations, then use the worked good-vs-weak answers to see exactly how marks are won and lost. Tick off the revision checklist at the end before a SAC or exam.

Study summary

A local network has walls. A global network has borders.

A network kept inside one building has a clear edge: a handful of doors you can lock and watch. The instant that network reaches across the internet to users, cloud servers and services in other countries, the edge disappears. You are now exposed to **everyone, everywhere, all the time** — and you are now answerable to the laws of **more than one country**.

The VCAA study design asks you to discuss the risks and benefits of using networks in a global environment. Don't reduce this to "hackers are bad." Examiners want you to see four distinct risk fronts that *specifically* get worse once a network goes global — and to weigh them honestly against the very real benefits that pushed the organisation global in the first place.

A corner shop closes at 6pm, knows its regulars, and only has to obey local council rules. Put that shop online and open it to the world: it never closes, anyone on Earth can walk in, you can be robbed while you sleep by someone you'll never identify, and you suddenly have to obey the consumer and privacy laws of every country your customers live in. Same shop, radically bigger risk surface — and a much bigger market.

Meet PixelPort

PixelPort is an indie-game storefront and community built by Aanya, a Year 12 student in Melbourne. Players around the world create accounts, download games, post on forums and buy in-game items.

- **Players** in Australia, the Philippines, Brazil, Germany and the USA — accounts, profiles, chat, ages and locations.

- **Game files & the database** live on a **cloud provider with data centres in Singapore and Virginia (USA)** — chosen because it's cheap and fast.
- **Payments** are handled by an **overseas payment company**, so card details travel offshore.
- **A global CDN** (content delivery network) copies game files to servers worldwide so downloads are quick everywhere.

Every convenience in that list is also a doorway. We'll use PixelPort to make each risk concrete — and to ask the real exam question: *is going global worth it?*

The four risk fronts of a global network

These are the four risk areas named for this dotpoint. Tap a card to flip it and see how it bites PixelPort specifically.

When a single packet crosses a border

Follow one player's sign-up. The moment their data leaves Australia, it falls under another country's rules — and Australian privacy law still follows it. Watch the packet travel from Melbourne to the overseas data centre.

This is the heart of the privacy and compliance risk. Under the Australian Privacy Principles , APP 8 (cross-border disclosure) says that before you send personal information overseas you must take reasonable steps to ensure the overseas recipient handles it the way Australian law requires — and you can stay accountable for what happens to it there. APP 11 still requires you to keep it secure. So "the cloud provider's data centre is overseas" is never an excuse: the obligation travels with the data.

Risks vs benefits — weigh them honestly

The dotpoint says risks **and benefits**. A top-band answer never lists only the scary half. Aanya didn't go global to be reckless — she went global because the upside is enormous. Examiners reward students who can hold both sides and then *make a judgement*.

- **Global market.** A bedroom project can reach millions of players in dozens of countries — impossible from a local-only network.
- **24/7 availability.** The store never closes; someone is always able to buy and play.
- **Cheap, scalable cloud.** Pay only for what you use; handle a launch-day spike without owning servers.
- **Speed & reliability via CDN.** Files are copied worldwide, so downloads are fast and one server failing doesn't take the store down.
- **Access to global services & talent.** Overseas payment, hosting and collaborators are a click away.
- **Bigger attack surface.** Reachable by every attacker on Earth, all day, every day.
- **Loss of data control.** Personal data sits in countries with different (sometimes weaker) protections.
- **Tangled compliance.** Must satisfy several countries' laws at once; mistakes mean heavy fines.
- **Harder to investigate.** Attackers in other jurisdictions are difficult to trace and prosecute.
- **Dependency on others.** An overseas provider's outage or breach instantly becomes *your* outage or breach.

Sort the incident

Five headlines hit PixelPort. For each, pick which of the four risk fronts it best illustrates. Instant feedback explains why.

P·E·L writing trainer

Most short-answer marks are won with a simple spine: make a **Point**, ground it in an **Example** (use PixelPort), then **Link** back to the question. Try it below — the tracker lights up as your answer hits each move.

"Explain **one** data-privacy risk that arises because PixelPort stores player data on overseas servers, and describe **one** strategy to reduce it." (3 marks)

VCE-style practice questions

Each question shows a weak and a strong response so you can see exactly what separates the bands. Read the marking guide, then mark your own attempt — your score builds in the dock.

The five things to walk into the exam with

- **Four risk fronts:** cyber security threats · data privacy · legal compliance · unauthorised access — each gets *worse* when a network goes global.
- **Always weigh benefits too:** global market, 24/7 uptime, scalable cloud, CDN speed/resilience, global services.
- **Cause → effect → global twist** is the recipe for a full-mark risk explanation.
- **Offshore ≠ off the hook:** Australian privacy duties (APP 8 cross-border disclosure, APP 11 security) follow the data overseas.
- **Finish with a judgement** on *discuss/evaluate* questions — ideally a conditional one ("stay global *provided* …").

Key terms

Term	Meaning
Global network	A network whose users, services and data span multiple countries, connected via the internet.
Attack surface	The total set of points where an attacker could try to get in — vastly larger once a system is reachable worldwide.
Data sovereignty	The idea that data is subject to the laws of the country it is physically stored in.
Cross-border disclosure (APP 8)	Sending personal data overseas; the sender must take reasonable steps to ensure it's handled under Australian standards.
DDoS	Distributed denial-of-service — many machines flood a server so real users can't get through.
Credential stuffing	Using passwords leaked from one site to break into reused accounts on another.
Man-in-the-middle	An attacker secretly intercepts data travelling between two points (e.g. between distant servers).

Term	Meaning
CDN	Content delivery network — copies of files held on servers worldwide so downloads are fast and resilient.
Notifiable Data Breaches scheme	Australian law (under the Privacy Act 1988) requiring eligible breaches of personal data to be reported.

Common traps & misconceptions

Exam trap
⚠ Common misconception Students write "the data is in another country so Australian law doesn't apply." **Wrong.** If an Australian organisation collects the data, Australian privacy obligations follow it across the border. Going global *adds* legal duties, it doesn't remove them.

Exam trap
① Naming a threat ≠ explaining the risk "DDoS is a risk" earns little. Say **what** happens and **why global makes it worse**: "a globally-distributed botnet floods the login server, and because attackers are overseas they're hard to trace." Cause → effect → the global twist.

Exam trap
② Forgetting the benefits If the command word is **discuss / evaluate** and you only list risks, you've answered half the question. Always weigh both sides, then judge.

Exam trap
③ "Overseas = Australian law doesn't apply" The opposite is true. Storing data offshore **adds** obligations (APP 8 cross-border disclosure) — it never cancels Australian privacy law.

Exam trap
④ Confusing "threat" with "unauthorised access" A DDoS is a **threat/attack**; an account takeover is **unauthorised access**. Answer the front the question actually names.

Exam trap
⑤ A strategy with no link back If asked for a mitigation, name it *and* link it to the specific risk: "multi-factor authentication stops credential stuffing because a stolen password alone is no longer enough to log in."

How to answer exam questions

Read the **command word** first — it sets how much to write. *State/Identify* wants a word or phrase; *Describe* wants features; *Explain* wants reasons and links; *Justify/Evaluate* wants a judgement backed by evidence. Always pull in the **scenario detail** rather than answering in the abstract. The examples below contrast a weak response with a strong one for the same question.

Q1. Explain · 3 marks

A global retailer stores customer data on servers in several countries. Explain two **risks** this creates for the organisation.

✗ A weak answer

Hackers might steal the data and that would be bad. *(→ One vague risk; no second risk and no legal dimension — 1.)*

✓ A strong answer

First, a wider **attack surface and unauthorised access**: more servers in more locations means more entry points for attackers to exploit, increasing the chance of a breach. Second, **legal-compliance risk**: different countries have different data-protection laws, so data lawful in one location may breach privacy legislation in another, exposing the retailer to penalties. Both threaten customer trust and the organisation's finances.

How the marks are awarded

- 1 mark — a valid technical risk (unauthorised access / larger attack surface).
- 1 mark — a valid legal/privacy-compliance risk.
- 1 mark — links a risk to a consequence for the organisation.

Q2. Describe · 2 marks

Describe why **data privacy** is a particular concern when data crosses international borders.

✗ A weak answer

Because other countries might read it. *(→ Too vague; no mention of differing laws — 0–1.)*

✓ A strong answer

When data crosses borders it becomes subject to the laws of each country it is stored or processed in. A country may have weaker privacy protections, or may legally compel access to the data, so personal information that is protected under Australian law could be exposed or used in ways the individual never consented to.

How the marks are awarded

- 1 mark — recognises data is subject to other jurisdictions' laws.
- 1 mark — links that to weaker protection / loss of control over personal data.

Revision checklist

- Four risk fronts: cyber security threats · data privacy · legal compliance · unauthorised access — each gets worse when a network goes global.
- Always weigh benefits too: global market, 24/7 uptime, scalable cloud, CDN speed/resilience, global services.
- Cause → effect → global twist is the recipe for a full-mark risk explanation.

- Offshore ≠ off the hook: Australian privacy duties (APP 8 cross-border disclosure, APP 11 security) follow the data overseas.
- Finish with a judgement on discuss/evaluate questions — ideally a conditional one ("stay global provided ...").